-- REPORT --

Replication of Förster, J., Liberman, N., & Kuschel, S. (2008).
The effect of global versus local processing styles on assimilation versus contrast in social judgment, *Journal of Personality and Social Psychology, 94,* 579-599.

Replication Author

David Reinhard
University of Virginia
reinhard@email.virginia.edu

Introduction

After priming a concept (such as "aggressive"), people can assimilate that concept into their social judgments (e.g. rate a person as being more aggressive compared to a no prime condition), or contrast their judgment away from the prime (e.g. rate the person as being less aggressive compared to a no prime condition). The paper by Förster, Liberman, and Kuschel (2008) proposed a "...Global/local processing style model (GLOMO) for assimilation and contrast effects in social judgment. According to GLOMO, inclusion versus exclusion should be influenced by whether people process information globally or locally."

In this replication of Förster et al. (2008), we sought to directly reproduce the methodology used in the original paper. The present work was conducted as part of the Reproducibility Project: Psychology, which typically asks volunteers to replicate the final study in their chosen paper. In this case, that would be Study 5. Study 5 relies on comparisons of students' drug and alcohol consumption to that of celebrities; pretesting, however, suggested that the students in the sampled population estimated that their drug and alcohol consumption was so low that the original procedure could not be followed. Therefore, the present replication is a replication of Study 1.

In Study 1, Förster et al. assessed whether or not cognitive processing style influenced the effect of priming on social judgments toward a target. To induce global and local processing, the authors first asked participants to view a map and attend to the overall shape (global processing), the smaller details (local processing), or both (control). Participants were then primed using aggressive (vs. neutral) words. Finally, participants read a text about a target, John, who displayed ambiguously aggressive behavior, and rated how aggressive the behavior was. The target effect for the replication is the interaction between cognitive processing style and priming on participants' ratings of John's behavior, $F(2, 76) = 21.57$, $p < .0001$.

Methods

Power Analysis

A power analysis to calculate the necessary sample size was conducted. An effect size was calculated using Daniel Lakens's spreadsheet for calculating effect sizes (2013). This yielded $\eta_p^2 = .362$. Power was calculated using G*Power: The test family was "F tests," the statistical test was "ANOVA: Fixed effects, special, main effects and interactions," the type of power analysis was "A priori: Compute required sample size - given α, power, and effect size." G*Power converted the effect size to f = .7532583. This was entered, along with the error probability of .05, power of .95, numerator df of 2, and 6 as the number of groups. This yielded 31 participants necessary for 95% power.

Planned Sample

Based on the above power analysis, 31 participants were necessary to have sufficient power to detect a significant replication of the effect of interest. The original study recruited 88 participants (but ran 82), however, so we planned to include the same N. Participants were recruited via an undergraduate participant pool and were run in one experimental session.

Materials

The use of original materials, described below by Forster et al. (2008), was followed as closely as possible. Differences from the original materials are discussed below in the section "Differences from the Original Study".

"*Induction of processing style*. ...[A map] was projected on the screen of the lecture hall, and participants were asked to examine it for 3 min. In the global processing condition, participants were asked to look at the map as a whole so that they could answer later questions about the overall shape of the map. In the local processing condition, participants were asked to look at the details of the map so that they could answer later questions about them. In the control condition, participants were asked to focus on both details and shape...

"*Semantic priming*. ...We used two word-search puzzles, 19 letters by 17 letters, each with 12 words that were arranged horizontally and vertically. All 12 words were listed on the right-hand side of the letter matrix. Both puzzles included 6 words that were neutral with respect to aggression (board, radio, roof, clock, newspaper, bread). The neutral puzzle included another 6 neutral words (teapot, colorful, calendar, barge, chair, cup) and the aggression puzzle included 6 aggression-related words (violence, hatred, brutal, murder, attack, malicious)...

"For our main dependent variable….participants read a paragraph about John, who is described as performing a number of ambiguously aggressive behaviors (e.g., denying a salesman entry into the apartment, refusing to pay rent until the landlord repainted the apartment). Previous research has shown that this paragraph is ambiguous with respect to its

evaluative implications, as the target person may be perceived as either aggressive or assertive...Participants then rated the target person on 10 bipolar scales, which ranged from 1 to 9, with endpoints labeled with antagonistic traits. The endpoints of the 2 critical scales, which appeared in the fifth and the eighth positions on the list, were labeled assertive– hostile and determined– belligerent. These 2 scales appeared along with 8 irrelevant scales (e.g., with endpoints labeled self-confident–arrogant, cowardly– cautious). The order of the traits was constant," (Förster et al., 2008, p. 583).

Procedure

The procedure was carried out in a lecture hall at the University of Virginia. Participants were informed that they were going to engage in three separate tasks examining different psychological questions.

Each participant was issued five folders individually labeled "1" through "5". The first folder had general instructions about the study. Participants were told that each of the other folders had a different task, one concerning geographic abilities, one measuring cognitive performance, and one concerning person perception. To indicate that each task was distinct from the others, the materials were "printed in different fonts and on different types of paper," (Förster et al., 2008, p. 583).

After reading the general instructions in the first folder, participants were instructed to open the second folder and read the instructions. "The second folder contained the alleged geographic abilities task to induce a global versus local processing style. Participants were asked to focus on a map...that was projected on the screen of the lecture hall. One third of the participants were instructed to attend to the global shape, one third to attend to the local details, and one third was asked to attend to both the details and shape of the map. After 3 min of observing the map, the participants were asked to answer a questionnaire concerning the map…," (Förster et al., 2008, p. 583).

Next, participants were instructed to open the third folder. "The third folder contained the semantic priming task…About half of the participants received the puzzle containing aggression-related words, whereas the other half received a puzzle containing no aggression-related words. Participants had 2 min to do the task…" (Förster et al., 2008, p. 583). Next, participants were told to open the fourth folder. "[It] contained the third impression formation task…They were asked to read the description of John and to rate him on the 10 bipolar scale." Finally, participants were told to open the fifth and final folder. "A final set of questions examined participants' awareness of the actual purpose of the study…" (Förster et al., 2008, p. 583).

Analysis Plan

The key effect was the interaction between cognitive processing style and priming. To examine this effect, we planned to conduct a 3 (processing style: global vs. local vs. control) X 2 (semantic priming: aggression vs. neutral) between-participants analysis of variance (ANOVA) on the average of the two aggressiveness ratings. We expected the interaction to show assimilation in the control and global conditions: participants would give higher aggression ratings after aggression priming than after neutral priming. However, we also expected that a contrast effect will occur after local processing: we anticipated that participants would give higher aggression ratings after control priming than after aggression priming. We planned to exclude participants that did not fully complete the dependent variables as well as participants that accurately guessed the hypotheses of the experiment.

Differences from the Original Study

The original work by Förster et al. had participants view a map of Oldenburg, Germany. Because the replication was run in the United States instead of Germany, we will show participants a map (provided by the original authors) of South Carolina instead. Not providing a map of a state with which all participants were especially familiar was important—if participants had significant prior knowledge of the state this would likely impact the efficacy of the prime. In the original study, participants were recruited from and participated during their Introduction to Social Psychology class. In the present replication, participants still took the study in a lecture hall with dozens of other students, but it did not occur during class time. Finally, participants in the original study were from 52 different nations while the students in the replication study are primarily from the United States. All changes were endorsed by the first author of the original study.

Post Data Collection Methods and Addendum

Actual Sample

74 students from the University of Virginia took part in the study (42 women, 32 men). We excluded 3 participants that correctly guessed the true purpose of the study. Our final sample is therefore 71 participants (41 women, 30 men).

Results

Data preparation

Data preparation followed the analysis plan.

Confirmatory analysis

The analysis was conducted as specified in the analysis plan.

In accordance with the original paper, we averaged the two aggression items to create an aggressiveness score. While there was a significant correlation between the two aggression items in the original paper, there was no significant relationship in the replication (r = .168, p = .163). In contrast to the original findings, there was no significant difference in aggressiveness ratings between participants primed with aggression (M = 6.88, SD = 1.55) and those not primed with aggression (M = 7.33, SD = 1.33), $t(69)$ = -1.28, p = .206. There was a significant difference in aggressiveness ratings between the processing style conditions, participants in the global processing condition gave higher aggressiveness ratings (M = 7.72, SD = 1.19) than those in the control condition (M = 6.60, SD = 1.31) and the local condition (M = 6.77, SD = 1.64), $F(2, 68)$ = 4.67, p = .013. To examine the target effect, we conducted a 3 (processing style: global vs. local vs. control) X 2 (semantic priming: aggression vs. neutral) between-participants ANOVA. There was no significant interaction between processing style and priming, $F(2, 65)$ = .865, p = .426, partial η^2= 0.0259 (see Figure 1 and Table 1).

Figure 1. Mean Aggressiveness Ratings as a Function of Induced Processing Style and Semantic Priming.

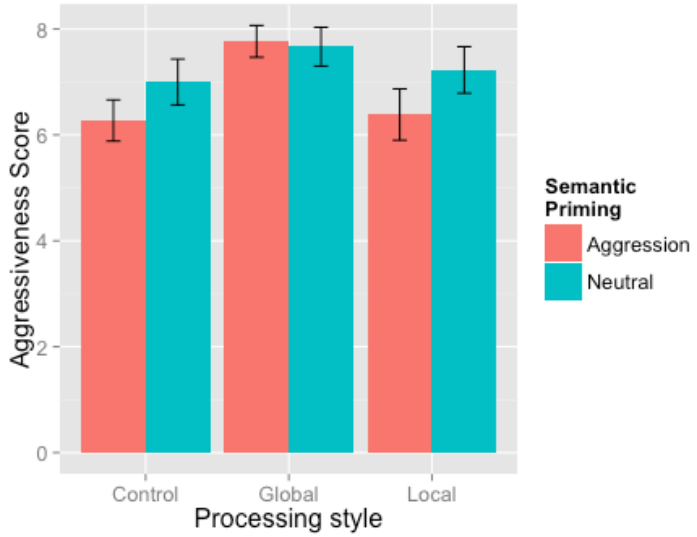


Table 1. Mean Aggressiveness Ratings (and Standard Deviations) as a Function of Induced Processing Style and Semantic Priming.

Induced Processing Style						
	Local		**Control**		**Global**	
	M	*SD*	*M*	*SD*	*M*	*SD*
Aggression	6.38	1.75	6.27	1.29	7.77	1.16
Neutral	7.23	1.46	7.00	1.30	7.67	1.27

Exploratory analyses

Because there was no significant relationship between the two aggression items, we ran the target interaction with each aggression item separately. For question 5 (Assertive-Hostile), there was no significant interaction between processing style and priming, $F(2, 65) = 1.11$, $p = .336$. For question 8 (Determined-Belligerent), there was also no significant interaction between processing style and priming, $F(2, 65) = .188$, $p = .829$.

We additionally checked the number of participants that reported using the correct cognitive processing style (e.g. participants that reported using a global focus in the global focus condition). Over half of the participants (38 out of 70, 1 participant did not provide a response) reported not using the instructed cognitive processing style. Therefore, we ran the target interaction using the participant's reported cognitive processing style. There was no significant interaction between reported processing style and priming, $F(2, 64) = 1.16$, $p = .321$.

Discussion

Summary of Replication Attempt

We replicated the main effect of processing style on aggression ratings, however the global processing induction (and not the local processing induction) mainly drove this effect. We did not replicate the main effect of semantic priming on aggression ratings. Finally, we did not replicate the target finding of a significant interaction between cognitive processing style and priming on participants' ratings of John's behavior.

Commentary

One possible explanation for the result could be the non-significant main effect of semantic priming, which is a necessary pre-condition for the interaction effect. Additionally, because the majority of participants reported using an incorrect processing style, the effectiveness of the processing style induction is unclear. The differences in stimuli between the original study and replication could have played a role in the effectiveness of the processing style induction. As such, these differences limit the conclusions about the lack of an interaction between processing style and semantic priming.

Before data collection, one of the authors from the original article speculated that there might be cultural differences in how politeness is interpreted between participants in the original sample (many who were from Eastern Europe) and our replication sample (who were primarily from the US). The original authors conducted unpublished pretests with students from the same university as those in the target study and did not replicate the basic aggressiveness effect (Higgins, Rholes, & Jones, 1977). Specifically, the effect requires participants rate the target's behavior as ambiguously aggressive; however, their participants did not believe the target

behaved aggressively enough. Therefore, they added more blatantly aggressive behaviors (e.g. “he grasped the arm of the store clerk”, see the “Final Materials” and “Original Materials” folder on OSF for all of the changes) to the text in order to replicate the original result. Because the mean aggression ratings were higher in the replication, it is possible that participants viewed the target’s behaviors as overtly aggressive instead of ambiguously aggressive. Also, because there was no significant relationship between the two averaged aggression items for the dependent variable, it is possible that students in the replication study interpreted the meaning of the aggression items differently than participants in the original study.

References

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